

with ASICs, which yield the greatest hash rate per dollar spent, while Ethereum is mined mostly with GPUs. Therefore, \$1,000 will purchase more hash rate for a Bitcoin computer than an Ethereum computer, and it is this dollar value that's most important in deterring attackers from attempting to recreate the network. Hence, while as of March 2017 Bitcoin's hash rate of 4,000,000 TH/s was technically 250,000-fold higher than Ethereum's 16,000 GH/s, this does not mean Bitcoin was 250,000 times more secure than Ethereum.

Decentralized Assets Should Have Decentralized Miners

Overall, hash rate is important, but so too is its decentralization. After all, if the hash rate is extremely high but 75 percent of it is controlled by a single entity, then that is not a decentralized system. It is actually a highly centralized system and therefore vulnerable to the whims of that one entity. If a cryptoasset is vulnerable to the whims of a single entity or small oligarchy, then that person or small group could choose to perform a 51 percent attack at some point, either to crush the value of the asset (a malicious kamikaze attack), or to try to profit from spending money they don't have. Such a risk must be considered and avoided.

Figures 13.3, 13.4, and 13.5 are charts showing the hash rate distribution among miners for Ethereum, Litecoin, and Bitcoin as of March 2017.